All commutators needed in CT theory

generated by `generate_commutators.py`

The commutators of the transcorrelated (CT) expansion, generated symbolically with sympy's `Commutator` and kept in nested-operator form. They are *not* expanded into concrete many-body equations; the expansion through the generalized Wick theorem and the cumulant expansion is done separately.

The truncation is by perturbation order (Møller–Plesset counting $\hat{F} \sim O(1)$, $\hat{V} \sim O(\lambda)$, $\hat{T} \sim O(\lambda)$), keeping the expansion correct through $O(\lambda^2)$: the full first-order term $[\hat{H}_b, \hat{A}]$ ($O(\lambda)+O(\lambda^2)$) and the Fock half of the second-order term, $\frac{1}{2}[[\hat{F}, \hat{A}], \hat{A}]$ ($O(\lambda^2)$). The valence double commutator $\frac{1}{2}[[\hat{V}, \hat{A}], \hat{A}]$ is $O(\lambda^3)$ and is not generated.

1 Elementary operators

The elementary operators have a direct many-body form (convention of `ct_F_A.tex`, `ct_V_A.tex` and `ct_f_a_a.pdf`: crossed Fock order, excitation operators, general indices over $\{i\} \cup \{a\}$):

$$\hat{h} = h^q_p \, \hat{E}^p_q, \qquad \hat{V} = \tfrac{1}{2} g^{pq}_{rs} \, \hat{E}^{rs}_{pq}, \tag{1}$$

$$\hat{F} = f^q_p \, \hat{E}^p_q, \qquad \hat{T}_1 = t^a_i \, \hat{E}^a_i, \tag{2}$$

$$\hat{T}_2 = \tfrac{1}{2} t^{ab}_{ij} \, \hat{E}^{ab}_{ij}, \tag{3}$$

with $\hat{h}$, $\hat{V}$ and $\hat{F}$ Hermitian.

2 Composite operators

The composite operators are sums and products of the elementary ones:

$$\hat{H}_b = \hat{h} + \hat{V}, \qquad \hat{T} = \hat{T}_1 + \hat{T}_2, \tag{4}$$

$$\hat{A} = \hat{T} - \hat{T}^\dagger, \qquad \bar{\hat{H}} = \hat{H}_b + \left[\hat{H}_b, \, \hat{A}\right] + \tfrac{1}{2}\left[\left[\hat{F}, \, \hat{A}\right], \, \hat{A}\right], \tag{5}$$

with $\hat{A}$ anti-Hermitian. The transcorrelated Hamiltonian $\bar{\hat{H}}$ is the target operator. The blocks of the first-order commutators are

$$\hat{B}_1 = \left[\hat{h}, \, \hat{T}_1\right], \qquad \hat{B}_2 = \left[\hat{h}, \, \hat{T}_2\right], \tag{6}$$

$$\hat{D}_1 = \left[\hat{V}, \, \hat{T}_1\right], \qquad \hat{D}_2 = \left[\hat{V}, \, \hat{T}_2\right], \tag{7}$$

$$\hat{C}_1 = \left[\hat{F}, \, \hat{T}_1\right], \qquad \hat{C}_2 = \left[\hat{F}, \, \hat{T}_2\right], \tag{8}$$

so that $[\hat{h}, \hat{A}] = \hat{B}_1 + \hat{B}_2 + \hat{B}_1^\dagger + \hat{B}_2^\dagger$, and similarly for $[\hat{V}, \hat{A}]$ and $[\hat{F}, \hat{A}]$.

3 First-order commutators

The first-order term of $\bar{\hat{H}}$ involves the composite bare Hamiltonian $\hat{H}_b = \hat{h} + \hat{V}$,

$$\left[\hat{H}_b,\, \hat{A}\right] = \left[\hat{h},\, \hat{A}\right] + \left[\hat{V},\, \hat{A}\right]. \tag{9}$$

Since $\hat{A} = \hat{T} - \hat{T}^\dagger$ and $[\hat{X}, \hat{T}]^\dagger = -[\hat{X}, \hat{T}^\dagger]$ for Hermitian $\hat{X}$, only the two $\hat{T}$ blocks have to be computed for each elementary part of $\hat{H}_b$:

$$\left[\hat{h},\, \hat{A}\right] = \left[\hat{h},\, \hat{T}_1\right] + \left[\hat{h},\, \hat{T}_2\right] + \text{h.c.}, \tag{10}$$

$$\left[\hat{V},\, \hat{A}\right] = \left[\hat{V},\, \hat{T}_1\right] + \left[\hat{V},\, \hat{T}_2\right] + \text{h.c.}. \tag{11}$$

4 Second-order commutators

At $O(\lambda^2)$ the second-order term keeps only the Fock half, $\frac{1}{2}[[\hat{F}, \hat{A}], \hat{A}]$; the valence half $\frac{1}{2}[[\hat{V}, \hat{A}], \hat{A}] \sim O(\lambda^3)$ is dropped. $\hat{A}$ anti-Hermitian makes $[\hat{F}, \hat{A}]$ Hermitian, so that $[[\hat{F}, \hat{A}], \hat{T}]^\dagger = -[[\hat{F}, \hat{A}], \hat{T}^\dagger]$ and

$$\frac{1}{2}\left[\left[\hat{F},\, \hat{A}\right],\, \hat{A}\right] = \tfrac{1}{2}\left[\left[\hat{F},\, \hat{A}\right],\, \hat{T}\right] + \text{h.c.}, \tag{12}$$

Expanding $[[\hat{F}, \hat{A}], \hat{T}] = [\hat{C}_1, \hat{T}] + [\hat{C}_2, \hat{T}] + [\hat{C}_1^\dagger, \hat{T}] + [\hat{C}_2^\dagger, \hat{T}]$ and each $[\,\cdot\,, \hat{T}] = [\,\cdot\,, \hat{T}_1] + [\,\cdot\,, \hat{T}_2]$ gives the eight blocks

$$\left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_1\right] \tag{13}$$

$$\left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_2\right] \tag{14}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_1\right] \tag{15}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_2\right] \tag{16}$$

$$\left[\left[\hat{F},\, \hat{T}_1\right]^\dagger,\, \hat{T}_1\right] \tag{17}$$

$$\left[\left[\hat{F},\, \hat{T}_1\right]^\dagger,\, \hat{T}_2\right] \tag{18}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_1\right] \tag{19}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_2\right] \tag{20}$$

The mixed nested commutators are different operators, $\left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_2\right] \neq \left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_1\right]$, since $\hat{T}_1\hat{T}_2 - \hat{T}_2\hat{T}_1$ does not vanish; the Jacobi shortcut is not available and both mixed blocks are kept.

5 Expansion of the target operator

Assembling the composite pieces and expanding every commutator of composite operators into commutators between elementary operators, the transcorrelated Hamiltonian through $O(\lambda^2)$ is

$$\begin{aligned}\bar{\hat{H}} &= \hat{H}_b + \left[\hat{H}_b,\, \hat{A}\right] + \frac{1}{2}\left[\left[\hat{F},\, \hat{A}\right],\, \hat{A}\right] \\ &= \hat{h} + \hat{V} + \left[\hat{h},\, \hat{T}_1\right] + \left[\hat{h},\, \hat{T}_2\right] + \left[\hat{V},\, \hat{T}_1\right] + \left[\hat{V},\, \hat{T}_2\right] + \text{h.c.} \\ &\quad + \tfrac{1}{2}\Big(\left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_1\right] + \left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_2\right] + \left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_1\right] + \left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_2\right] \\ &\qquad + \left[\left[\hat{F},\, \hat{T}_1\right]^\dagger,\, \hat{T}_1\right] + \left[\left[\hat{F},\, \hat{T}_1\right]^\dagger,\, \hat{T}_2\right] + \left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_1\right] + \left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_2\right] + \text{h.c.}\Big).\end{aligned} \tag{21}$$

Every remaining object is a commutator between elementary operators ($\hat{h}$, $\hat{V}$, $\hat{F}$ with $\hat{T}_1$, $\hat{T}_2$); these are the quantities that the generalized Wick theorem and the cumulant expansion turn into many-body equations.

6 Complete list

The commutators that remain to be evaluated (each by one or two applications of the $(1,1)$, $(1,2)$, $(2,1)$ or $(2,2)$ rule), as returned by `commutator_blocks()`:

First order (4 blocks + h.c.)

$$\left[\hat{h},\, \hat{T}_1\right] \tag{22}$$

$$\left[\hat{h},\, \hat{T}_2\right] \tag{23}$$

$$\left[\hat{V},\, \hat{T}_1\right] \tag{24}$$

$$\left[\hat{V},\, \hat{T}_2\right] \tag{25}$$

Second order (8 blocks + h.c.)

$$\left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_1\right] \tag{26}$$

$$\left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_2\right] \tag{27}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_1\right] \tag{28}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_2\right] \tag{29}$$

$$\left[\left[\hat{F},\, \hat{T}_1\right]^\dagger,\, \hat{T}_1\right] \tag{30}$$

$$\left[\left[\hat{F},\, \hat{T}_1\right]^\dagger,\, \hat{T}_2\right] \tag{31}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_1\right] \tag{32}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_2\right] \tag{33}$$